

Sri Lanka Institute of Information Technology



Lab Submission

IT25104032 – Kahandawa K.A.O

Lab 02 - Student Management System using Classes, Packages & Control Structures

Object Oriented Analysis and Design | SE2012

B.Sc. (Hons) Computer Science

Parts 01-04

Code:

1. Student.java

```
package models;

/**
 * Represents a student with a student ID and name.
 */

public class Student {
    private String id;
    private String name;

    /**
     * Creates a Student object.
     *
     * @param id the student's ID
     * @param name the student's name
     */

    public Student(String id, String name) {
        this.id = id;
        this.name = name;
    }

    /**
     * Returns the student ID.
     *
     * @return the student ID
     */
    public String getId() {
        return id;
    }

    /**
     * Returns the student name.
     *
     * @return the student name
     */
}
```

```
        */
    public String getName() {
        return name;
    }

    /**
     * Returns the student's details as a formatted
    string.
     *
     * @return formatted student information
     */
    public String toString() {
        return ("Student ID: " + id + "\tStudent Name:
    " + name);
    }
}
```

2. Course.java

```
package models;

/**
 * Represents a course with a course code and title.
 */

public class Course {
    private String code;
    private String title;

    /**
     * Creates a Course object.
     *
     * @param code the course code
     * @param title the course title
     */
    public Course(String code, String title) {
        this.code = code;
        this.title = title;
    }

    /**
     * Returns the course code.
     *
     * @return the course code
     */
    public String getCode() {
        return code;
    }

    /**
     * Returns the course title.
     *
     * @return the course title
     */
    public String getTitle() {
        return title;
    }
}
```

```
    }

    /**
     * Returns the course details as a formatted
string.
     *
     * @return formatted course information
    */
    public String toString() {
        return ("Course Code: " + code + "\tCourse
Title: " + title);
    }

}
```

3. EnrollmentManager.java

```
package services;

import models.Student;

/**
 * Manages student related operations such as searching
 * and sorting.
 */

public class EnrollmentManager {

    /**
     * Creates an EnrollmentManager object.
     */
    public EnrollmentManager() {

    }

    /**
     * Searches for a student using the student ID.
     *
     * @param students the array containing student
     * objects
     * @param count the number of students currently
     * stored
     * @param id the student ID to search for
     * @return the matching student if found; otherwise
     * null
     */
    public Student searchStudentById(Student[]
students, int count, String id) {
        for (int i = 0; i < count; i++) {
            if
(students[i].getId().equalsIgnoreCase(id)) {
                return students[i];
            }
        }
        return null;
    }
}
```

```

    }

    /**
     * Sorts students alphabetically by name using
Bubble Sort.
     *
     * @param students the array of students
     * @param count the number of students currently
stored
     */
    public void sortStudentsByName(Student[] students,
int count) {

        for (int i = 0; i < count - 1; i++) {
            for (int j = 0; j < count - i - 1; j++) {
                if
(students[j].getName().compareToIgnoreCase(students[j +
1].getName())) > 0) {
                    Student temp = students[j];
                    students[j] = students[j + 1];
                    students[j + 1] = temp;
                }
            }
        }
    }
}

```

4. Main.java

```
package mainapp;

import models.Student;
import services.EnrollmentManager;

import java.util.Scanner;

/**
 * Runs the Student Management System.
 * Handles user interaction through a menu-driven
 * interface.
 */

public class Main {

    /**
     * Starts the Student Management System.
     *
     * @param args command-line arguments (not used)
     */
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        Student[] students = new Student[100];
        int studentCount = 0;

        EnrollmentManager manager = new
        EnrollmentManager();

        int option;

        do {
            System.out.println("MAIN MENU");
            System.out.println("1. Enroll student");
            System.out.println("2. List students");
            System.out.println("3. Search student");
            System.out.println("4. Sort students");
```



```

        System.out.println("5. Exit");

        System.out.print("Enter your choice: ");
        option = input.nextInt();
        input.nextLine();

        switch(option) {
            case 1:
                if (studentCount < 100) {
                    System.out.print("Enter
student's name: ");
                    String name =
input.nextLine();
                    System.out.print("Enter
student ID: ");
                    String id = input.nextLine();
                    students[studentCount] = new
Student(id, name);
                    studentCount++;
                    System.out.println("Student
enrolled successfully");
                }else {
                    System.out.println("Can't
enroll more students");
                }
                break;

            case 2:
                if (studentCount == 0) {
                    System.out.println("No
students available");
                }else {
                    System.out.println("----
Student details ----");
                    for (int i = 0; i <
studentCount; i++) {

                        System.out.println(students[i]);
                    }
                }
            }
        }
    }
}

```

```

        }
        break;

    case 3:
        System.out.print("Enter student
ID: ");

        String searchID =
input.nextLine();

        Student studentval =
manager.searchStudentById(students, studentCount,
searchID);

        if (studentval == null) {
            System.out.println("Student
not found");

        }else {

            System.out.println(studentval);
        }
        break;

    case 4:

        manager.sortStudentsByName(students, studentCount);
        System.out.println("Students
sorted successfully");
        break;

    case 5:
        System.out.println("Exiting...");
        break;

    default:
        System.out.println("Invalid
option");

        break;
    }
}while (option != 5);

```

```

        input.close();
    }
}

```

Output:

```

Windows PowerShell
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PS D:\Onadi\SLIIT\Year2\Y2S1\Modules - Y2S1\SE2012 - Object Oriented Analysis and Design\Labs\Lab02> git clone https://github.com/IT25104032/SE2012-2026-S2-L02.git
Cloning into 'SE2012-2026-S2-L02'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 2 (from 1)
Receiving objects: 100% (3/3), done.
PS D:\Onadi\SLIIT\Year2\Y2S1\Modules - Y2S1\SE2012 - Object Oriented Analysis and Design\Labs\Lab02> cd .\SE2012-2026-S2-L02\
PS D:\Onadi\SLIIT\Year2\Y2S1\Modules - Y2S1\SE2012 - Object Oriented Analysis and Design\Labs\Lab02\SE2012-2026-S2-L02> cd .\StudentManagement\
PS D:\Onadi\SLIIT\Year2\Y2S1\Modules - Y2S1\SE2012 - Object Oriented Analysis and Design\Labs\Lab02\SE2012-2026-S2-L02\StudentManagement> javac models\Student.java models\Course.java services\EnrollmentManager.java mainapp\Main.java
PS D:\Onadi\SLIIT\Year2\Y2S1\Modules - Y2S1\SE2012 - Object Oriented Analysis and Design\Labs\Lab02\SE2012-2026-S2-L02\StudentManagement> java mainapp.Main
MAIN MENU
1. Enroll student
2. List students
3. Search student
4. Sort students
5. Exit
Enter your choice: 1
Enter student's name: Melissa
Enter student ID: 032
Student enrolled successfully
MAIN MENU
1. Enroll student
2. List students
3. Search student
4. Sort students
5. Exit
Enter your choice: 1
Enter student's name: Sheldon
Enter student ID: 035
Student enrolled successfully

```

```

MAIN MENU
1. Enroll student
2. List students
3. Search student
4. Sort students
5. Exit
Enter your choice: 1
Enter student's name: Sheldon
Enter student ID: 035
Student enrolled successfully
MAIN MENU
1. Enroll student
2. List students
3. Search student
4. Sort students
5. Exit
Enter your choice: 1
Enter student's name: Georgie
Enter student ID: 040
Student enrolled successfully
MAIN MENU
1. Enroll student
2. List students
3. Search student
4. Sort students
5. Exit
Enter your choice: 2
----- Student details -----
Student ID: 032 Student Name: Melissa
Student ID: 035 Student Name: Sheldon
Student ID: 040 Student Name: Georgie
MAIN MENU
1. Enroll student
2. List students
3. Search student
4. Sort students
5. Exit
Enter your choice: 3
Enter student ID: 035
Student ID: 035 Student Name: Sheldon

```

```

MAIN MENU
1. Enroll student
2. List students
3. Search student
4. Sort students
5. Exit
Enter your choice: 3
Enter student ID: 066
Student not found
MAIN MENU
1. Enroll student
2. List students
3. Search student
4. Sort students
5. Exit
Enter your choice: 4
Students sorted successfully
MAIN MENU
1. Enroll student
2. List students
3. Search student
4. Sort students
5. Exit
Enter your choice: 2
---- Student details ----
Student ID: 040 Student Name: Georgie
Student ID: 032 Student Name: Melissa
Student ID: 035 Student Name: Sheldon
MAIN MENU
1. Enroll student
2. List students
3. Search student
4. Sort students
5. Exit
Enter your choice: 5
Exiting...

```

Parts 05 – Generate JavaDoc

```

PS D:\Onadi\SLIIT\Year2\Y2S1\Modules - Y2S1\SE2012 - Object Oriented Analysis and Design\Labs\Lab02\SE2012-2026-S2-L02\StudentManagement> javadoc -d docs mo
dels/*.java services/*.java
Loading source file models\Course.java...
Loading source file models\Student.java...
Loading source file services\EnrollmentManager.java...
Constructing Javadoc information...
Creating destination directory: "docs\"
Building index for all the packages and classes...
Standard Doclet version 21.0.8+12-LTS-250
Building tree for all the packages and classes...
Generating docs\models\Course.html...
Generating docs\services\EnrollmentManager.html...
Generating docs\models\Student.html...
Generating docs\models\package-summary.html...
Generating docs\models\package-tree.html...
Generating docs\services\package-summary.html...
Generating docs\services\package-tree.html...
Generating docs\overview-tree.html...
Generating docs\index.html...
Building index for all classes...
Generating docs\allclasses-index.html...
Generating docs\allpackages-index.html...
Generating docs\index-all.html...
Generating docs\search.html...
Generating docs\overview-summary.html...
Generating docs\help-doc.html...

```